

PL/SQL:

Exercise 1: Control Structures

```
1 DECLARE
2   -- Customer variables
3   TYPE customer_rec IS RECORD (
4     CustomerID NUMBER,
5     Name VARCHAR2(50),
6     Age NUMBER,
7     Balance NUMBER,
8     ISVIP VARCHAR2(5)
9   );
10
11   TYPE customer_tab IS TABLE OF customer_rec INDEX BY PLS_INTEGER;
12   customers customer_tab;
13
14   -- Loan variables
15   TYPE loan_rec IS RECORD (
16     CustomerID NUMBER,
17     InterestRate NUMBER,
18     DueInDays NUMBER
19   );
20
21   TYPE loan_tab IS TABLE OF loan_rec INDEX BY PLS_INTEGER;
22   loans loan_tab;
23
24 BEGIN
25   -- Sample Customer Data
26   customers(1).CustomerID := 1;
27   customers(1).Name := 'Rahul';
28   customers(1).Age := 65;
29   customers(1).Balance := 12000;
30   customers(1).ISVIP := 'FALSE';
31
32   customers(2).CustomerID := 2;
33   customers(2).Name := 'Priya';
34   customers(2).Age := 45;
35   customers(2).Balance := 8000;
```

```
36   customers(2).Balance := 8000;
37   customers(2).ISVIP := 'FALSE';
38
39   customers(3).CustomerID := 3;
40   customers(3).Name := 'Amit';
41   customers(3).Age := 70;
42   customers(3).Balance := 15000;
43   customers(3).ISVIP := 'FALSE';
44
45   -- Loan Data
46   loans(1).CustomerID := 1;
47   loans(1).InterestRate := 10;
48   loans(1).DueInDays := 20;
49
50   loans(2).CustomerID := 2;
51   loans(2).InterestRate := 9;
52   loans(2).DueInDays := 45;
53
54   loans(3).CustomerID := 3;
55   loans(3).InterestRate := 11;
56   loans(3).DueInDays := 10;
57
58   DBMS_OUTPUT.PUT_LINE('----- Scenario 1 -----');
59
60   FOR i IN 1..customers.COUNT LOOP
61     IF customers(i).Age > 60 THEN
62       FOR j IN 1..loans.COUNT LOOP
63         IF loans(j).CustomerID = customers(i).CustomerID THEN
64           loans(j).InterestRate := loans(j).InterestRate - 1;
65           DBMS_OUTPUT.PUT_LINE(customers(i).Name ||
66             ' New Interest Rate = ' || loans(j).InterestRate || '%');
67         END IF;
68       END LOOP;
69     END IF;
70   END LOOP;
```

```
71   END IF;
72 END LOOP;
73
74 DBMS_OUTPUT.PUT_LINE(' ');
75 DBMS_OUTPUT.PUT_LINE('----- Scenario 2 -----');
76
77 FOR i IN 1..customers.COUNT LOOP
78   IF customers(i).Balance > 10000 THEN
79     customers(i).ISVIP := 'TRUE';
80     DBMS_OUTPUT.PUT_LINE(customers(i).Name || ' promoted to VIP');
81   END IF;
82 END LOOP;
83
84 DBMS_OUTPUT.PUT_LINE(' ');
85 DBMS_OUTPUT.PUT_LINE('----- Scenario 3 -----');
86
87 FOR i IN 1..customers.COUNT LOOP
88   FOR j IN 1..loans.COUNT LOOP
89     IF customers(i).CustomerID = loans(j).CustomerID
90       AND loans(j).DueInDays <= 30 THEN
91       DBMS_OUTPUT.PUT_LINE('Reminder: Dear ' ||
92         customers(i).Name ||
93         ', your loan is due in ' ||
94         loans(j).DueInDays || ' days. ');
95     END IF;
96   END LOOP;
97 END LOOP;
98
99 END;
```

STDM
Input for the program (Optional)

Output:

```
----- Scenario 1 -----
Rahul New Interest Rate = 9%
Amit New Interest Rate = 10%
----- Scenario 2 -----
Rahul promoted to VIP
Amit promoted to VIP
----- Scenario 3 -----
Reminder: Dear Rahul, your loan is due in 20 days.
Reminder: Dear Amit, your loan is due in 10 days.
```

STDM
Input for the program (Optional)

Output:

```
----- Scenario 1 -----
Rahul New Interest Rate = 9%
Amit New Interest Rate = 10%
----- Scenario 2 -----
Rahul promoted to VIP
Amit promoted to VIP
----- Scenario 3 -----
Reminder: Dear Rahul, your loan is due in 20 days.
Reminder: Dear Amit, your loan is due in 10 days.
```

Light W W Oracle PL/SQL

Exercise 3: Stored Procedures

```
1 DECLARE
2
3 -- Scenario 1: Process Monthly Interest
4 PROCEDURE ProcessMonthlyInterest IS
5     balance NUMBER := 10000;
6     interestRate NUMBER := 1;
7 BEGIN
8     balance := balance + (balance * interestRate / 100);
9
10    DBMS_OUTPUT.PUT_LINE('----- Scenario 1 -----');
11    DBMS_OUTPUT.PUT_LINE('Interest Applied: 1%');
12    DBMS_OUTPUT.PUT_LINE('Updated Balance: ' || balance);
13 END;
14
15 -- Scenario 2: Update Employee Bonus
16 PROCEDURE UpdateEmployeeBonus(
17     dept VARCHAR2,
18     bonusPercent NUMBER
19 ) IS
20     salary NUMBER := 50000;
21 BEGIN
22     salary := salary + (salary * bonusPercent / 100);
23
24    DBMS_OUTPUT.PUT_LINE(chr(10) || '----- Scenario 2 -----');
25    DBMS_OUTPUT.PUT_LINE('Department: ' || dept);
26    DBMS_OUTPUT.PUT_LINE('Bonus Percentage: ' || bonusPercent || '%');
27    DBMS_OUTPUT.PUT_LINE('Updated Salary: ' || salary);
28 END;
29
30 -- Scenario 3: Transfer Funds
31 PROCEDURE TransferFunds(
32     fromBalance NUMBER,
33     toBalance NUMBER,
34     amount NUMBER
35 ) IS
36
37     sender NUMBER := fromBalance;
38     receiver NUMBER := toBalance;
39 BEGIN
40     DBMS_OUTPUT.PUT_LINE(chr(10) || '----- Scenario 3 -----');
41
42     IF sender >= amount THEN
43         sender := sender - amount;
44         receiver := receiver + amount;
45
46         DBMS_OUTPUT.PUT_LINE('Transfer Successful');
47         DBMS_OUTPUT.PUT_LINE('Sender Balance: ' || sender);
48         DBMS_OUTPUT.PUT_LINE('Receiver Balance: ' || receiver);
49     ELSE
50         DBMS_OUTPUT.PUT_LINE('Transfer Failed: Insufficient Balance');
51     END IF;
52 END;
53
54 -- Calling all procedures
55 BEGIN
56     ProcessMonthlyInterest;
57
58     UpdateEmployeeBonus('IT',10);
59
60     TransferFunds(10000,5000,1000);
61 END;
62 /
```

STDM
Input for the program (optional)

Output:

----- Scenario 1 -----
Interest Applied: 1%
Updated Balance: 10100

----- Scenario 2 -----
Department: IT
Bonus Percentage: 10%
Updated Salary: 55000

----- Scenario 3 -----
Transfer Successful
Sender Balance: 7000
Receiver Balance: 8000

----- Scenario 3 -----
Transfer Successful
Sender Balance: 7000
Receiver Balance: 8000